

Laboratory Manual

PHP with Laravel (MCA29503)

Program: MCA 2nd Semester | Session: 2023-24

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Assignment 1 — Basic PHP

1.1 Hello World

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Hello World PHP</title>
</head>
<body>
<?php
  echo "Hello World!";
?>
</body>
</html>
```

Output:

Hello World!

1.2 HTML Form with PHP Retrieve

HTML Form Code:

```
<form action="gre3.php" method="post">
  Enter your Name: <input type="text" name="name">
  Enter Your Phone No. <input type="text" name="phone">
  <input type="submit" value="Submit">
</form>
```

PHP Retrieve Code:

```
<?php
```

```
echo "Name: " . $_POST["name"];  
echo "Phone: " . $_POST["phone"];  
?>
```

Output:

```
Welcome  
Name: John  
Phone No.: 9876543210
```

1.3 Swap Two Numbers

Code:

```
<?php  
$number1 = $_POST["number1"];  
$number2 = $_POST["number2"];  
echo "Before Swap: 1st=$number1, 2nd=$number2";  
$number1 = $number1 + $number2;  
$number2 = $number1 - $number2;  
$number1 = $number1 - $number2;  
echo "After Swap: 1st=$number1, 2nd=$number2";  
?>
```

Output (Input: 10, 20):

```
Before Swap: 1st=10, 2nd=20  
After Swap: 1st=20, 2nd=10
```

1.4 Odd or Even

Code:

```
<?php  
$number = $_POST["number"];  
if($number % 2 == 0){  
    echo "Enter number is $number that number is Even";  
} else {  
    echo "Enter number is $number that number is Odd";  
}  
?>
```

Output:

```
Enter number is 7 that number is Odd  
Enter number is 4 that number is Even
```

Assignment 2 — Control Structures

2.1 Maximum of Three Numbers

Code:

```
<?php
    $n1=$_POST["number1"]; $n2=$_POST["number2"]; $n3=$_POST["number3"];
    if($n1 > $n2){
        if($n1 > $n3) echo "Maximum num $n1";
        else echo "Maximum num $n3";
    } else {
        if($n3 > $n2) echo "Maximum num $n3";
        else echo "Maximum num $n2";
    }
?>
```

Output (Input: 10, 25, 18):

Numbers are 10, 25, 18
Maximum num 25

2.2 Multiplication Table

Code:

```
<?php
    $number1 = $_POST["number1"];
    for($i=1; $i<11; $i++){
        $mul = $i * $number1;
        echo "$i X $number1 = $mul<br>";
    }
?>
```

Output (Input: 5):

1 X 5 = 5
2 X 5 = 10
3 X 5 = 15
4 X 5 = 20
5 X 5 = 25
... up to 10 X 5 = 50

2.3 Chess Board using For Loop

Code:

```
<?php
for($row=1; $row<=8; $row++){
    echo "<tr>";
    for($column=1; $column<=8; $column++){
        $total = $row + $column;
        if($total % 2 == 0)
```

```

        echo "<td bgcolor=#FFFFFF></td>";
    else
        echo "<td bgcolor=#000000></td>";
    }
    echo "</tr>";
}
?>

```

Output:

An 8x8 chess board with alternating black and white cells is rendered in the browser.

Assignment 3 — Calculations

3.1 Electricity Bill Calculator

Code:

```

<?php
function calculate_bill($units) {
    if($units <= 50) {
        $bill = $units * 3.50;
    } else if($units <= 100) {
        $bill = (50*3.5) + (($units-50)*4.00);
    } else if($units <= 200) {
        $bill = (50*3.5) + (100*4.0) + (($units-150)*5.20);
    } else {
        $bill = (50*3.5) + (100*4.0) + (100*5.20) + (($units-250)*6.50);
    }
    return number_format((float)$bill, 2, ".", "");
}
$result = calculate_bill(120);
echo "Total amount for 120 units: Rs. " . $result;
?>

```

Output (Input: 120 units):

Total amount of 120 - 747.00

3.2 Mathematical Calculator

Code:

```

<?php
if(isset($_POST["submit"])){
    $fnumber = $_POST["fnumber"];
    $snumber = $_POST["snumber"];
}

```

```

$choice = $_POST["choice"];
switch($choice){
    case "ADD": $res=$fnumber+$snumber; $msg="Addition"; break;
    case "SUB": $res=$fnumber-$snumber; $msg="Subtraction"; break;
    case "MUL": $res=$fnumber*$snumber; $msg="Multiplication"; break;
    case "DIV": $res=$fnumber/$snumber; $msg="Division"; break;
}
echo "$msg of $fnumber, $snumber is $res";
}
?>

```

Output (Input: 10, 5, ADDITION):

Addition of 10,5 is 15

Assignment 4 — String Manipulation

4.1 String Operations

Input String:

"the quick brown FOX jumps over the lazy DOG"

Code:

```

<?php
$inputstring = "the quick brown FOX jumps over the lazy DOG";
echo strtoupper($inputstring); // Lowercase to Uppercase
echo strtolower($inputstring); // Uppercase to Lowercase
echo ucfirst($inputstring);    // First char uppercase
echo ucwords($inputstring);    // All words first char uppercase
echo substr($inputstring, -11); // Last 11 characters
$words = explode(" ", $inputstring);
echo $words[1];                // Second word
?>

```

Output:

Lowercase→Uppercase: THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG

Uppercase→Lowercase: the quick brown fox jumps over the lazy dog

First char uppercase: The quick brown FOX jumps over the lazy DOG

All words uppercase: The Quick Brown Fox Jumps Over The Lazy Dog

Last 11 characters: the lazy DOG

Replace 'the' → 'best': best quick brown FOX jumps over the lazy DOG

Second word: quick

4.2 Count Vowels in String

Code:

```
<?php
function countVowels($string){
    $vowels = array("a","e","i","o","u");
    $len = strlen($string); $num=0;
    for($i=0; $i<$len; $i++){
        if(in_array($string[$i], $vowels)){
            echo "$string[$i] - position is $i<br>";
            $num++;
        }
    }
    return $num;
}
$word = strtolower($_POST["word"]);
echo "$word - have No. of Vowels are " . countVowels($word);
?>
```

Output (Input: madam):

```
a - position is 1
a - position is 3
madam - have No. of Vowels are 2
```

Assignment 5 — Built-in Functions

5.1 Palindrome Check

Code:

```
<?php
function isPalindrome($number) {
    $numberStr = strval($number);
    $reverseStr = strrev($numberStr);
    return $numberStr === $reverseStr;
}
$givenNumber = 12321;
if (isPalindrome($givenNumber))
    echo $givenNumber . " is a palindrome.";
else
    echo $givenNumber . " is not a palindrome.";
?>
```

Output:

```
12321 is a palindrome.
```

(For 12345) → 12345 is not a palindrome.

5.2 Display Current Date using getdate()

Code:

```
<?php
$date_array = getdate();
foreach($date_array as $key => $val){
    echo $key . "=" . $val . "<br>";
}
echo "<h1>Today's date: " . $date_array["mon"] . "/" .
    $date_array["mday"] . "/" . $date_array["year"] . "</h1>";
?>
```

Output:

```
seconds=30 minutes=45 hours=14 mday=10 wday=3
mon=3 year=2024 yday=69 weekday=Wednesday month=March ...
Today's date: 3/10/2024
```

5.3 Digital Clock

Code:

```
<script>
function showTime(){
    var d = new Date();
    var h = d.getHours(); var m = d.getMinutes(); var s = d.getSeconds();
    h = (h<10)?"0"+h:h; m = (m<10)?"0"+m:m; s = (s<10)?"0"+s:s;
    document.getElementById("clock").innerHTML = h+":"+m+":"+s;
    setTimeout(showTime, 1000);
}
showTime();
</script>
<div id="clock" style="font-size:48px; font-family:monospace;"></div>
```

Output:

```
14:35:22 (Digital clock updating every second)
```

5.4 Get Server Information

Code:

```
<?php
echo "<pre>";
print_r($_SERVER);
echo "</pre>";
?>
```

Output:

```
Array (
  [SERVER_NAME] => localhost
  [SERVER_PORT] => 80
  [DOCUMENT_ROOT] => C:/xampp/htdocs
  [REQUEST_METHOD] => GET
  [PHP_SELF] => /progrm.php
  ... (all server variables displayed)
)
```

Assignment 6 — Arrays

6.1 Array Operations

Array: \$countries =

['Chile','Colombia','India','India','Italy','Chile','Austria','New Zealand','United States']

Code (key operations):

```
<?php
$countries = array("Chile","Colombia","India","India","Italy",
                  "Chile","Austria","New Zealand","United States");

// 1. Display using foreach
foreach($countries as $value) echo "<br>".$value;

// 2. Sorted Array
sort($countries);
foreach($countries as $value) echo "<br>".$value;

// 3. Without duplicates
$uarray = array_unique($countries);
foreach($uarray as $value) echo "<br>".$value;

// 4. Remove last element
array_pop($countries);
foreach($countries as $value) echo "<br>".$value;

// 5. Reverse
$revarr = array_reverse($countries);
foreach($revarr as $value) echo "<br>".$value;
```

```
// 6. Insert at position 2
array_splice($countries, 2, 0, "Germany");

// 7. Search element
if(in_array("India", $countries))
    echo "Found at: ".array_search("India", $countries);
?>
```

Output:

1. Display: Chile, Colombia, India, India, Italy, Chile, Austria, New Zealand, United States
2. Sorted: Austria, Chile, Chile, Colombia, India, India, Italy, New Zealand, United States
3. No Duplicates: Chile, Colombia, India, Italy, Austria, New Zealand, United States
4. Remove Last: Chile, Colombia, India, India, Italy, Chile, Austria, New Zealand
5. Reverse: New Zealand, Austria, Chile, Italy, India, India, Colombia, Chile
6. After Insert 'Germany' at pos 2: Chile, Colombia, Germany, India...
7. Search 'India': Found Search element India at position 2

Assignment 7 — Array Key Operations

7.1 Get Array Key from Array List

Code:

```
<?php
$array = array("apple","orange","banana","kiwi","lemon");
$search = $_POST["search"];
print_r(array_keys($array, $search));
?>
```

Output (Search: banana):

```
Array ( [0] => 2 )
```

Assignment 8 — Form Handling

8.1 Self-Processing Form (GET/POST)

Code:

```
<?php if ($_SERVER["REQUEST_METHOD"] === "GET") : ?>
<form action="" method="post">
```

```

    Name: <input type="text" name="name" required />
    Email: <input type="email" name="email" required />
    <button type="submit">Subscribe</button>
</form>
<?php else : ?>
<?php
    $name = htmlspecialchars($_POST["name"]);
    $email = htmlspecialchars($_POST["email"]);
    echo "Thanks $name for your subscription.";
    echo "Please confirm it in your inbox: $email";
    ?>
<?php endif ?>

```

Output (after POST):

Thanks John for your subscription.
Please confirm it in your inbox of the email john@example.com.

8.2 Task Form with List Box

HTML Code:

```

<form action="process_form.php" method="post">
    Task Name: <input type="text" name="taskName" required>
    Message: <textarea name="message" rows="4"></textarea>
    Task: <select name="task">
        <option>Create Task</option>
        <option>Update Task</option>
        <option>Delete Task</option>
    </select>
    <input type="submit" value="Add">
</form>

```

PHP Code:

```

<?php
$taskName = $_POST["taskName"];
$message = $_POST["message"];
$selectedTask = $_POST["task"];
echo "<h2>Task Information</h2>";
echo "<p><strong>Task Name:</strong> $taskName</p>";
echo "<p><strong>Message:</strong> $message</p>";
echo "<p><strong>Selected Task:</strong> $selectedTask</p>";
?>

```

Output:

Task Information
Task Name: My First Task
Message: This is a test task

Selected Task: Create Task

Assignment 9 — Form Validation

9.1 CV Registration Form with Validation

HTML (Client-Side Validation):

```
<form action="process_registration.php" method="post" onsubmit="return validateForm()">
  Full Name: <input type="text" id="fullName" name="fullName" required>
  Email: <input type="email" id="email" name="email" required>
  Phone: <input type="tel" id="phone" name="phone" pattern="[0-9]{10}"
required>
  Address: <textarea id="address" name="address" rows="4" required></textarea>
  Resume URL: <input type="url" id="resume" name="resume" required>
  <input type="submit" value="Submit">
</form>
```

PHP Server-Side Code:

```
<?php
if($_SERVER["REQUEST_METHOD"] == "POST"){
    $fullName = $_POST["fullName"];
    $email = $_POST["email"];
    $phone = $_POST["phone"];
    $errors = [];
    if(empty($fullName)) $errors[] = "Full Name is required.";
    if(!filter_var($email, FILTER_VALIDATE_EMAIL)) $errors[] = "Invalid
email.";
    if(!preg_match("/^\d{10}$/", $phone)) $errors[] = "Invalid phone.";
    if(!empty($errors)) {
        foreach($errors as $e) echo "<p>$e</p>";
    } else {
        echo "<h2>Registration Successful</h2>";
        echo "<p>Full Name: $fullName</p>";
        echo "<p>Email: $email</p>";
    }
}
?>
```

Output (Success):

```
Registration Successful
Full Name: John Doe
Email: john@example.com
Phone: 9876543210
```

Address: 123, Park Street, Kolkata
Resume URL: <https://johndoe.com/resume>

Output (Validation Error):

Full Name is required.
Invalid email format.
Invalid phone number (10 digits only).

Assignment 10 — File Handling

10.1 Get Directory, Filename & Line Number

Code:

```
<?php
echo "Current directory: " . __DIR__ . "<br>";
echo "Current file: " . __FILE__ . "<br>";
echo "Line number: " . __LINE__ . "<br>";
?>
```

Output:

Current directory: C:/xampp/htdocs/phplab
Current file: C:/xampp/htdocs/phplab/progrm.php
Line number: 4

10.2 Read from Directory

Code:

```
<?php
$current_dir = __DIR__;
$dir = opendir($current_dir);
echo "<p>Upload directory is $current_dir</p>";
echo "<ul>";
while($file = readdir($dir))
    echo "<li>" . $file . "</li>";
echo "</ul>";
closedir($dir);
?>
```

Output:

Upload directory is C:/xampp/htdocs/phplab
Directory Listing: ., .., index.php, progrm.php, upload/ ...

10.3 File Upload with Info

Code:

```
<?php
if($_FILES){
    print_r($_FILES);
    echo "Filename: " . $_FILES["file"]["name"];
    echo "Type: " . $_FILES["file"]["type"];
    echo "Size: " . $_FILES["file"]["size"];
    echo "Temp name: " . $_FILES["file"]["tmp_name"];
    echo "Error: " . $_FILES["file"]["error"];
}
?>
```

Output (after upload):

```
Filename: sample.pdf
Type: application/pdf
Size: 102400
Temp name: C:\Windows\Temp\phpABC.tmp
Error: 0
```

10.4 Upload Image to 'upload' Folder**Code:**

```
<?php
if(isset($_FILES["file"])){
    $file_name = $_FILES["file"]["name"];
    $file_ext = strtolower(end(explode(".", $file_name)));
    $extensions = array("jpeg", "jpg", "png");
    if(!in_array($file_ext, $extensions))
        $errors[] = "Only JPEG or PNG allowed.";
    if(file_exists("upload/" . $file_name))
        $errors[] = "File already exists.";
    if(empty($errors))
        move_uploaded_file($_FILES["file"]["tmp_name"], "upload/". $file_name);
    echo "Success";
}
?>
```

Output:

```
Success (file saved to /upload/photo.jpg)
(On error) → extension not allowed, please choose a JPEG or PNG file.
```

Assignment 11 — MySQL To-Do App

11.1 Create Table

SQL:

```
CREATE TABLE todo_list (  
  id          INT AUTO_INCREMENT PRIMARY KEY,  
  task_name   VARCHAR(255) NOT NULL,  
  task_description TEXT,  
  due_date   DATE,  
  status     ENUM("Pending","Completed") DEFAULT "Pending"  
);
```

11.2 PHP To-Do CRUD

PHP Code (Insert + View):

```
<?php  
$conn = new mysqli("localhost","root","","tododb");  
if($conn->connect_error) die("Connection failed.");  
  
// INSERT  
if(isset($_POST["addTask"])){  
  $sql = "INSERT INTO todo_list (task_name, task_description, due_date)",  
        " VALUES ('$taskName','$taskDesc','$dueDate')";  
  if($conn->query($sql)) echo "Task added successfully!";  
}  
  
// SELECT  
$result = $conn->query("SELECT * FROM todo_list");  
while($row = $result->fetch_assoc()){  
  echo $row["id"]." | ".$row["task_name"]." | ".$row["status"];  
}  
$conn->close();  
?>
```

Output:

Task added successfully!

```
-----  
ID | Task Name      | Status  
1 | Buy groceries  | Pending  
2 | Submit report  | Completed  
3 | Call dentist   | Pending
```

Assignment 12 — Laravel Setup

12.1 Install Composer & Create Laravel Project

Commands:

```
# Install Composer (download installer from getcomposer.org)

# Create new Laravel project
composer create-project --prefer-dist laravel/laravel myblog

# Navigate to project
cd myblog

# List Artisan commands
php artisan list

# Start development server
php artisan serve
```

Output:

```
Laravel development server started: http://127.0.0.1:8000
Server started. Browse to http://localhost:8000
```

12.2 Basic Blog App (Routes & Views)

routes/web.php:

```
use App\Http\Controllers\BlogController;

Route::get('/', [BlogController::class, 'home'])->name('home');
Route::get('/about', [BlogController::class, 'about'])->name('about');
Route::get('/blog', [BlogController::class, 'blog'])->name('blog');
```

BlogController.php:

```
<?php
namespace App\Http\Controllers;

class BlogController extends Controller {
    public function home() { return view('home'); }
    public function about() { return view('about'); }
    public function blog() { return view('blog'); }
}
```

resources/views/home.blade.php:

```
<!DOCTYPE html><html>
<head><title>My Blog</title></head>
<body>
    <h1>Welcome to My Blog</h1>
    <nav>
        <a href='{{ route('home') }}'>Home</a> |
```

```
<a href='{{ route('about') }}'>About</a> |  
<a href='{{ route('blog') }}'>Blog</a>  
</nav>  
</body></html>
```

Output:

Browser shows: Welcome to My Blog with navigation links Home | About | Blog

Assignment 13 — Laravel Controllers, Views & Middleware

13.1 To-Do App with Middleware

.env Configuration:

```
DB_CONNECTION=mysql  
DB_HOST=127.0.0.1  
DB_PORT=3306  
DB_DATABASE=todo_laravel  
DB_USERNAME=root  
DB_PASSWORD=  
  
API_KEY=mysecretapikey123
```

Create Middleware:

```
php artisan make:middleware CheckApiKey
```

app/Http/Middleware/CheckApiKey.php:

```
<?php  
namespace App\Http\Middleware;  
  
class CheckApiKey {  
    public function handle($request, Closure $next) {  
        if ($request->header('X-API-Key') !== env('API_KEY')) {  
            return response()->json(['error'=>'Unauthorized'], 401);  
        }  
        return $next($request);  
    }  
}
```

Apply Middleware in routes/web.php:

```
Route::middleware(['CheckApiKey'])->group(function () {  
    Route::resource('tasks', TaskController::class);  
});
```

Output:

http://localhost:8000/tasks → Shows task list (if API key valid)
Without API key → 401 Unauthorized JSON response

Assignment 14 — Eloquent, Sessions & Cookies

14.1 Eloquent Model & CRUD

Create Model & Migration:

```
php artisan make:model Task -m
```

database/migrations/..._create_tasks_table.php:

```
Schema::create('tasks', function (Blueprint $table) {  
    $table->id();  
    $table->string('task_name');  
    $table->text('task_description')->nullable();  
    $table->date('due_date')->nullable();  
    $table->enum('status', ['Pending','Completed'])->default('Pending');  
    $table->timestamps();  
});
```

app/Models/Task.php:

```
<?php  
namespace App\Models;  
use Illuminate\Database\Eloquent\Model;  
  
class Task extends Model {  
    protected $fillable = ['task_name','task_description','due_date','status'];  
}
```

TaskController.php (Eloquent CRUD):

```
// CREATE  
Task::create($request->all());  
  
// READ  
$tasks = Task::all();  
  
// UPDATE  
$task = Task::findOrFail($id);  
$task->update(['status' => 'Completed']);  
  
// DELETE  
Task::destroy($id);
```

Session & Cookie Usage:

```
// Store in session
```

```
session(['user_name' => 'John']);

// Retrieve from session
$name = session('user_name');

// Store cookie (expires in 60 minutes)
return response('Hello')->cookie('user', 'John', 60);

// Read cookie
$user = request()->cookie('user');
```

Output:

**Task list page shows all tasks from database with Pending/Completed status.
Login stores user_name in session; accessible across pages.
Cookie 'user' set and retrieved across requests.**